

# A Derivation of Wien's Law by Dimensional Analysis

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## Abstract

*The Wien displacement law is the capstone of classical physics.* – Wannier [1]

A derivation of Wien's law by dimensional analysis is presented. It suggests existence of a universal constant that relates frequency to energy allowing the construction of a dimensionless quantity from  $\omega/T$ . Planck's constant is indeed such a constant.

## Derivation

What we want to determine is the energy spectral density per unit volume,  $u_\omega$ , of electromagnetic wave in empty space in equilibrium with surrounding matter at temperature  $T$ . We know that temperature is proportional to the average energy of the system and that Boltzmann constant  $k$  relates temperature to energy. Since we want to find the spectral density of electromagnetic wave, let us use the speed of light  $c$  [m/s] and its angular frequency  $\omega$  [rad/s] as well as  $kT$  [J]. There is no other physical quantity to specify this system.

The Rayleigh–Jeans law is obtained by constructing the energy spectral density per unit volume [J/(rad/s)/m<sup>3</sup>] from these three quantities:

$$u_\omega = A \frac{kT}{c^3} \omega^2, \quad (1)$$

where  $A$  is a dimensionless constant. The spectral density,  $u_\omega$ , must satisfy Stefan's  $T^4$  law:

$$u(T) = \int_0^\infty u_\omega d\omega = a T^4. \quad (2)$$

Suppose that this dimensionless quantity is in fact a function of  $\omega$  and  $T$ . We can write such a function as  $A(\omega T^r)$ .<sup>1</sup> Inserting (1) into (2) yields

$$u(T) = \frac{kT}{c^3} \int_0^\infty \omega^2 A(\omega T^r) d\omega.$$

We want to take  $T$  out from the integral. Let  $y = \omega T^r$ , we obtain

$$u(T) = \frac{kT}{c^3} \int_0^\infty \frac{y^2}{T^{2r}} A(y) \frac{1}{T^r} dy = \frac{kT^{(1-3r)}}{c^3} \int_0^\infty y^2 A(y) dy.$$

Since the integral is constant,  $r$  must be  $-1$  to satisfy Stefan's law (2). Therefore,

$$u_\omega = A(\omega/T) \frac{kT}{c^3} \omega^2 = \omega^3 \frac{A(\omega/T)}{c^3} kT/\omega.$$

This is Wien's law. You can easily show that if  $u_\omega$  has a maximum at a frequency, the frequency is proportional to  $T$ .

How do we construct a dimensionless quantity from  $\omega/T$ ? We know that the Boltzmann constant relates temperature to energy. Therefore it suggests a universal constant which relates frequency to energy. Planck's constant is indeed such a constant.

## Acknowledgments

I acknowledge the use of ChatGPT to proofread and check the grammar of the text. This derivation is not my original. However, I lost the reference.

## References

- [1] G. H. Wannier, *Statistical Physics*. Dover, 1987.

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<sup>1</sup>In case  $A$ 's dependency is  $\omega^s T^r$ , we can always rewrite  $A$  as  $A((\omega T^{r/s})^s)$ , and replace  $A$  and  $r$  by  $A'(x) = A(x^s)$  and  $r' = r/s$ .